

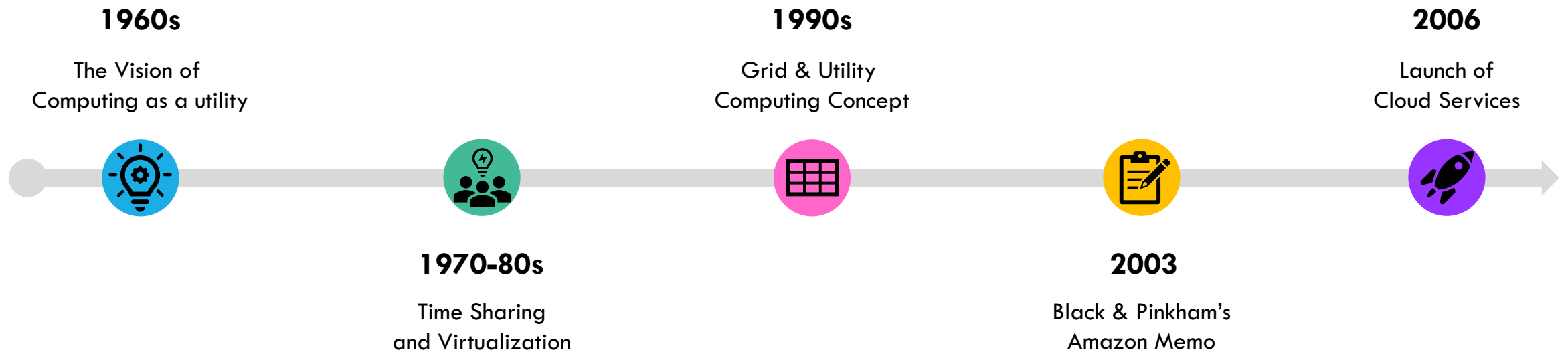
GETTING STARTED WITH RESEARCH COMPUTING IN THE CLOUD: AWS, GCP, AND AZURE

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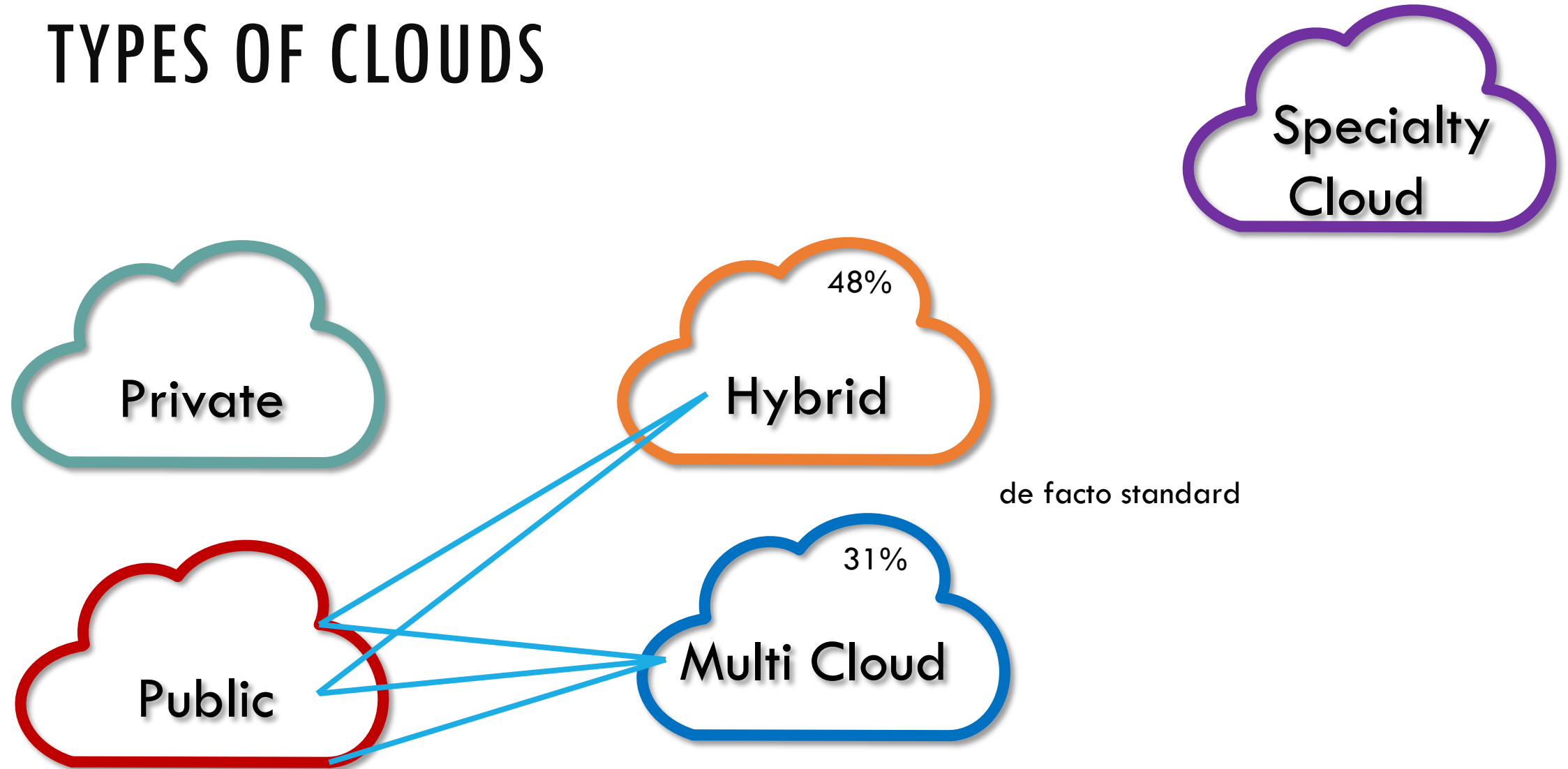


FORECASTING THE CLOUD

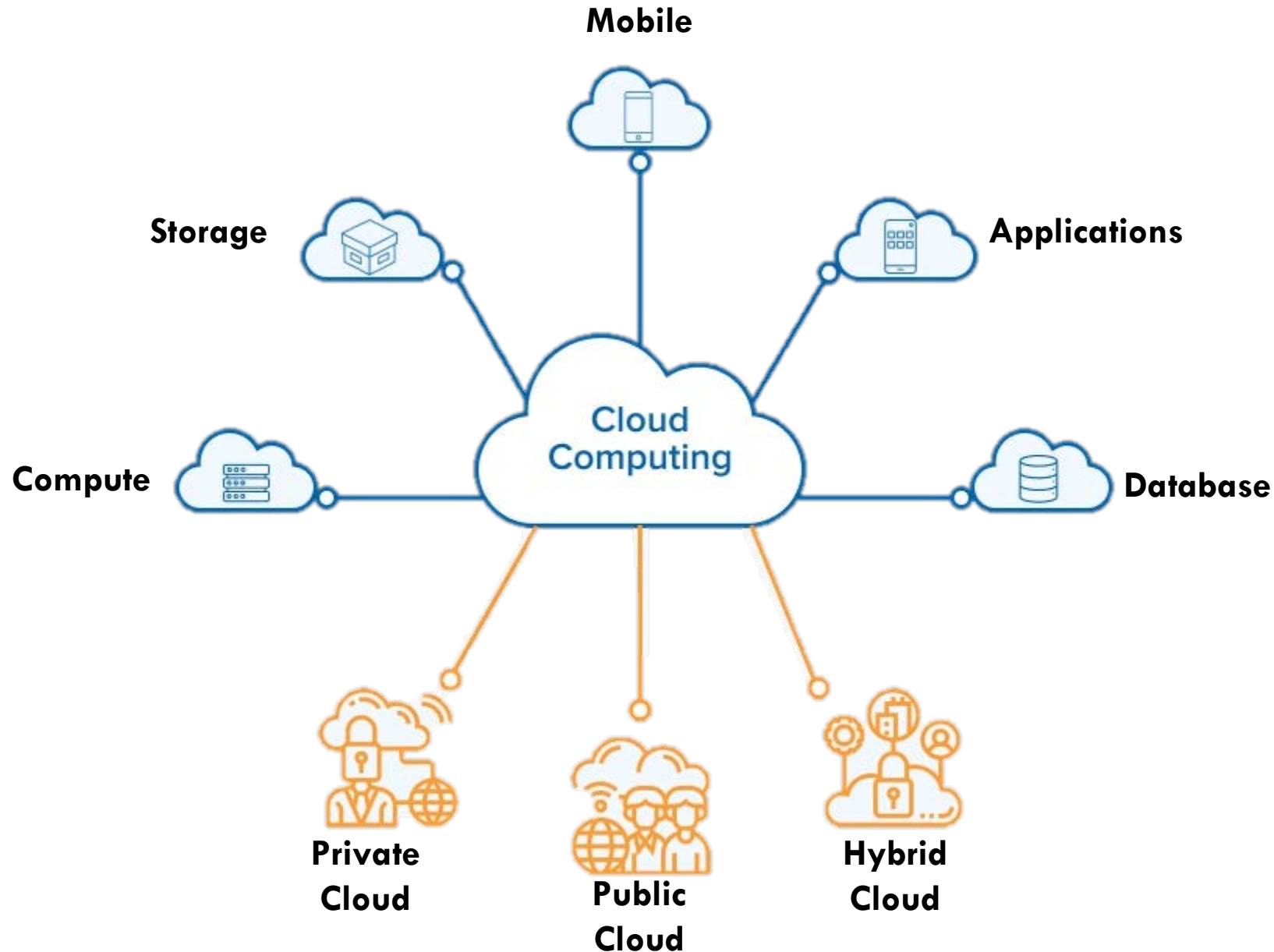


*"What's interesting [now] is that there is an emergent new model... It starts with the premise that the data services and architecture should be on servers. **We call it cloud computing.**"* - Eric Schmidt (then CEO of Google), 2006

TYPES OF CLOUDS



KEY CLOUD DEPLOYMENT COMPONENTS



WHY CLOUD?

Reduced cost

1

2

Flexibility

Scalability

3

4

Built-in backup plan

Technology options

5

6

Collaboration

AZURE

Pros

New market leader

Tight Microsoft products integration

Built on the existing customer base

Cons

Customer support issues

Partner ecosystem

Resource limitations



AWS

Pros

Market leader for 10 years

Massive Scope of operations

Comprehensive Global Network

Cons

Cost Structure

Support Fees

Overwhelming implementation choices



GCP

Pros

Strong Offer in Container

Specializes in High compute offerings

Strong AI/ML capabilities

Cons

Limited Offerings

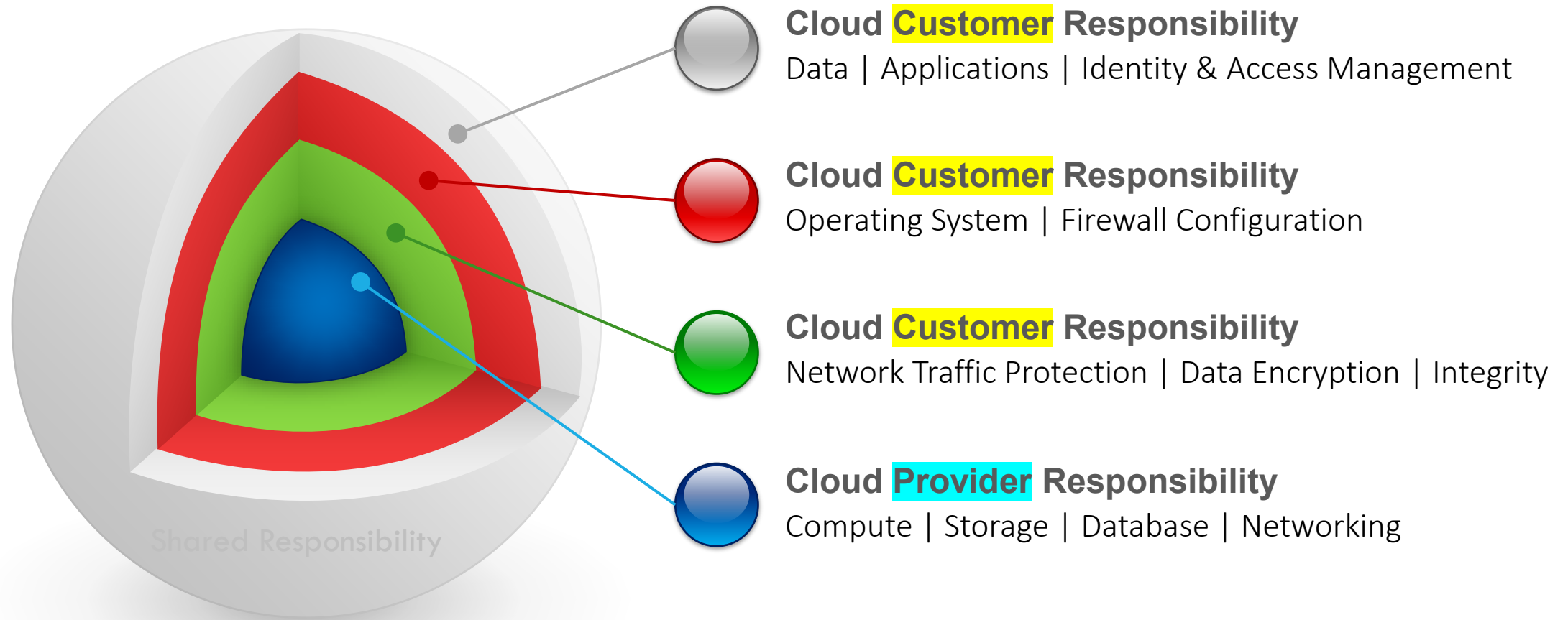
Limited Global Data Centers

Secondary provider, not a strategic provider

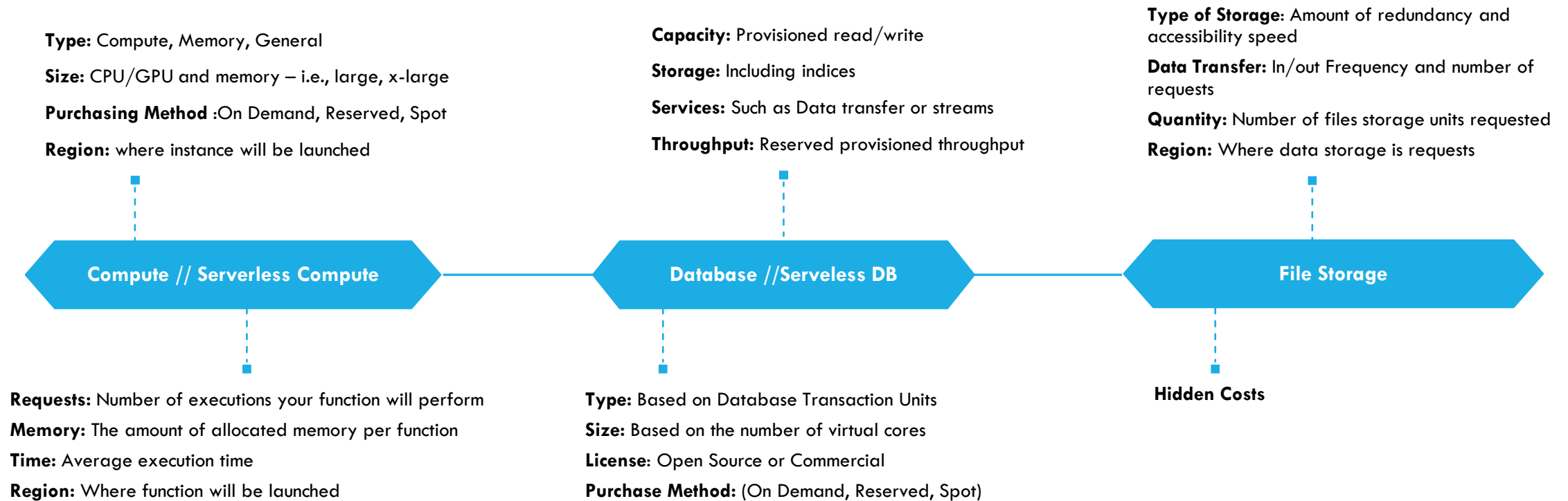


Google
Cloud Platform

CLOUD SECURITY RESPONSIBILITY



UNDERSTAND COST FACTORS



STRATEGIES FOR COST SAVINGS

RIGHT SIZE

Right Size the machine for the job



SHUTDOWN SCHEDULE

Control running hours for resources



SPRAWL CONTROL

Control sprawl of virtual machines



DISCOUNTS

If discount available, use it!



EXPIRATION DATE

Monitor & delete expired instances



TAGS

Keep track of resources with logical tags



STRATEGIES FOR COST SAVINGS

ORPHAN CHECK

Check for orphaned storage and lingering snapshots



LOGGING

Log user activities for audit and cross reference



ALERTS

Alerts and alarms notify you of reaching budget constraints



BUDGETS

Utilize budgets that will ensure you won't overspend



LIFECYCLE MGMT

Automatically move data to a less expensive storage or purge data



AUTOMATE

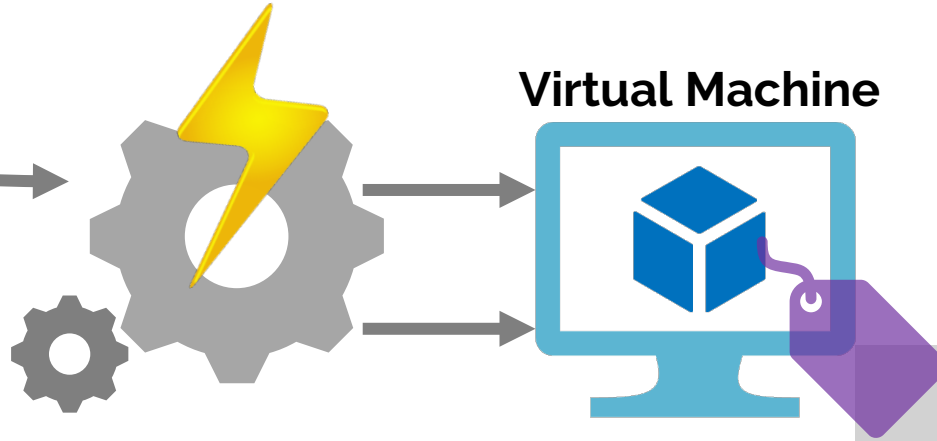
Use Templates to stand-up/bring down stacks



AUTOMATION EXAMPLE



Start VM



Shutdown VM



```
{  
  "Monday-Thursday": {  
    "Start": "9",  
    "Shutdown": "17"  
  },  
  "Friday": {  
    "Start": "9",  
    "Shutdown": "16"  
  },  
}
```

RESOURCES

Presenter



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Research Computing



<https://researchcomputing.princeton.edu/systems/cloud-computing>

OIT Cloud Accounts



https://princeton.service-now.com/service?id=sc_cat_item&table=sc_cat_item&sys_id=06268c7c1bc444d098d1217e6e4bcb4f

Pricing Calculators



<https://calculator.aws/#/>



<https://azure.microsoft.com/en-us/pricing/calculator/>



<https://cloud.google.com/products/calculator/>

Google Cloud

